PROJECT TEST PLAN

Project #22 - Team 28 - CODEVERSE

1. Document Control

Field	Value
Project	AI-Powered Upskilling Recommendation and Automation System for Students
Project ID	22
Deliverable	Project Test Plan
Team	Team 28 - CODEVERSE
Team SRNs	PES2UG24AM184, PES2UG24AM217, PES2UG24AM210, PES2UG24AM214
Version	1.0
Status	Planned test cases; Actual Result and Test Result to be updated after execution

2. Purpose and Scope

This test plan defines the verification approach for the implemented scope of Project #22. The system is a student-facing web application covering account/authentication, student profile, skill management, ML-based career recommendation, skill-gap analysis, learning roadmap and progress tracking. External job-portal automation, resume parsing, GitHub/LinkedIn integration, notifications and an admin portal are outside the current mini-project scope.

3. Test Objectives

• Verify every implemented functional module behaves according to its requirements.

• Cover Unit, Integration and System testing.

• Include both valid and invalid inputs and controlled error handling.

• Verify data flow among frontend, backend, database and ML recommendation components.

• Maintain traceable test IDs for later linkage to Jira stories and the requirements traceability matrix.

4. Test Levels

Level	Purpose	Examples
Unit Testing (UT)	Validate individual functions/classes/modules in isolation.	Validation rules, skill comparison, roadmap ordering, progress calculation.

Level	Purpose	Examples
Integration Testing (IT)	Validate interactions between modules/services/data stores.	API ↔ PostgreSQL, backend ↔ ML engine, recommendation ↔ roadmap.
System Testing (ST)	Validate complete user-facing workflows against expected behaviour.	Registration → profile → skills → recommendation → gap → roadmap → progress.

5. Test Environment

Component	Planned Environment
Client	Modern Chromium-based browser on Windows/Linux/macOS
Frontend	Project web frontend served locally or through the project deployment
Backend	Python-based REST API service
Database	PostgreSQL development/test database
ML component	Versioned trained recommendation model/service
Testing	pytest for Python unit/integration tests; browser/manual system testing
Tracking	Git + GitHub; Jira for backlog and defects
CI/CD	Jenkins as required by the course project guidelines

6. Test Data Strategy

• Use synthetic student accounts and non-sensitive sample profiles.

• Prepare beginner, intermediate and advanced skill profiles.

• Include complete, partial, duplicate, unknown and invalid skill inputs.

• Prepare career records with required skills and prerequisite relationships.

• Use controlled model versions/responses for repeatable recommendation tests.

• Do not use real passwords or unnecessary personal information.

7. Entry and Exit Criteria

Entry Criteria	Exit Criteria
Build is deployable in the test environment; database schema is available; required test data exists; APIs are reachable.	All planned cases are executed; Actual Result and Pass/Fail are populated; critical defects are fixed or formally recorded; traceability is updated; test summary is reviewed.

8. Defect Handling

Any failed test case should create a Jira defect containing the test case ID, steps to reproduce, expected result, actual result, severity, evidence and assigned owner. After the fix, the test should be re-run and its result updated.

9. Test Case Convention

UT = Unit Test, IT = Integration Test, ST = System Test. Each test case has a unique ID, module, description, preconditions, ordered steps, precise test data, expected result, actual result and Pass/Fail result. Actual Result and Test Result are intentionally marked as not executed/TBD until the team performs the required manual execution.

10. Detailed Test Cases

AUTH: Authentication & Account — Owner: PES2UG24AM184 — Requirements: FR-01 to FR-05

ID	Type	Test Case Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_AUTH_01	UT	Validate registration with valid student data	User is on registration screen	Enter valid name, email, password; submit	Name=Arun; email=arun@pesu.edu; password=Strong@123	Account is created and a success response is returned	Not executed — populate after manual test	TBD
UT_AUTH_02	UT	Reject duplicate email during registration	An account already exists for the email	Submit registration using existing email	email=existing@pesu.edu	Registration is rejected with a duplicate-email message	Not executed — populate after manual test	TBD
UT_AUTH_03	UT	Validate password policy	Registration form is available	Enter a password below minimum requirement; submit	password=123	Validation error is shown and account is not created	Not executed — populate after manual test	TBD
IT_AUTH_04	IT	Verify registration persists user data	Database is available	Register a new user; query user record	new valid user	User record is stored with required fields	Not executed — populate after manual test	TBD

ID	Type	Test Case Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
IT_AUTH_05	IT	Verify login after successful registration	A valid account exists	Enter registered email/password and submit	registered email + correct password	Login succeeds and authenticated session/token is issued	Not executed — populate after manual test	TBD
IT_AUTH_06	IT	Reject incorrect login credentials	A valid account exists	Enter correct email with wrong password	wrong password	Login fails without exposing sensitive account details	Not executed — populate after manual test	TBD
ST_AUTH_07	ST	Verify unauthenticated user cannot open protected dashboard	No active session	Open dashboard URL directly	/dashboard	User is redirected to login	Not executed — populate after manual test	TBD
ST_AUTH_08	ST	Verify logout terminates authenticated access	User is logged in	Click logout; reopen protected page	Valid session	Session is terminated and protected page is inaccessible	Not executed — populate after manual test	TBD

PROF: Student Profile — Owner: PES2UG24AM184 — Requirements: FR-06 to FR-09

ID	Type	Test Case Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_PROF_01	UT	Save a valid student profile	User is authenticated	Enter mandatory profile fields and save	Name, degree, year, interests	Profile validation passes and data is accepted	Not executed — populate after manual test	TBD
UT_PROF_02	UT	Reject invalid year value	User is authenticated	Enter unsupported year and save	year=9	Validation error is shown	Not executed — populate after manual test	TBD
UT_PROF_03	UT	Edit existing profile	Profile exists	Change degree/interests and save	Updated degree/interests	Updated values replace previous values	Not executed — populate after manual test	TBD
IT_PROF_04	IT	Persist profile updates to database	User is authenticated	Update profile; reload page	Updated profile	Updated data is returned from database	Not executed — populate after manual test	TBD
IT_PROF_05	IT	Retrieve profile for recommendation request	Profile and skills exist	Open recommendation page	Authenticated user	Current profile data is available to recommendation service	Not executed — populate after manual test	TBD
IT_PROF_06	IT	Prevent another user from editing profile	Two user accounts exist	Attempt update using another user's identifier	User A token + User B ID	Request is denied	Not executed — populate after manual test	TBD
ST_PROF_07	ST	Display profile correctly after relogin	Profile has been saved	Logout, login again, open profile	Valid account	Saved profile is displayed accurately	Not executed — populate after manual test	TBD
ST_PROF_08	ST	Handle incomplete profile gracefully	Authenticated user has missing optional fields	Open recommendation workflow	Partial profile	System explains missing required inputs rather than failing	Not executed — populate after manual test	TBD

SKILL: Skill Management — Owner: PES2UG24AM217 — Requirements: FR-10 to FR-13

ID	Type	Test Case Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_SKILL_01	UT	Add a valid skill	User is authenticated	Select skill and proficiency; save	Python, Intermediate	Skill is added to profile	Not executed — populate after manual test	TBD
UT_SKILL_02	UT	Reject invalid proficiency value	User is authenticated	Submit unsupported proficiency	proficiency=Expert++	Validation rejects value	Not executed — populate after manual test	TBD
UT_SKILL_03	UT	Prevent duplicate skill entry	Skill already exists	Add same skill again	Python	Duplicate is prevented or existing skill is updated	Not executed — populate after manual test	TBD
IT_SKILL_04	IT	Persist skill and proficiency	User is authenticated	Add skill; refresh profile	Python, Advanced	Skill remains after reload	Not executed — populate after manual test	TBD

ID	Type	Test Case Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
IT_SKILL_05	IT	Delete an existing skill	Skill exists	Click delete; confirm	Python	Skill is removed from profile	Not executed — populate after manual test	TBD
IT_SKILL_06	IT	Use updated skills in recommendation request	Skills exist	Change skill level; request recommendations	Python=Advanced	Recommendation service receives updated skill data	Not executed — populate after manual test	TBD
ST_SKILL_07	ST	Display skill list accurately	Multiple skills exist	Open skills section	Python, SQL, Git	All saved skills and proficiency levels are shown	Not executed — populate after manual test	TBD
ST_SKILL_08	ST	Handle empty skill profile	User has no skills	Open recommendation flow	No skills	System requests skill input instead of producing an invalid recommendation	Not executed — populate after manual test	TBD

REC: Career Recommendation — Owner: PES2UG24AM217 — Requirements: FR-14 to FR-18

ID	Type	Test Case Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_REC_01	UT	Generate recommendation for a valid profile	Profile and skills are complete	Submit recommendation request	CS student + Python, SQL, Git	One or more ranked career paths are returned	Not executed — populate after manual test	TBD
UT_REC_02	UT	Return deterministic result for same input	Same profile and model version	Request recommendation twice	Identical profile	Results are consistent within defined ranking tolerance	Not executed — populate after manual test	TBD
UT_REC_03	UT	Handle unsupported / unknown skill	Authenticated user	Request recommendation with unknown skill	Skill=XYZUnknown	Unknown skill is ignored/flagged without crashing	Not executed — populate after manual test	TBD
IT_REC_04	IT	Pass profile, skills and model features correctly	Backend and ML service available	Trigger recommendation	Valid student profile	Correct feature vector is sent to model	Not executed — populate after manual test	TBD
IT_REC_05	IT	Store recommendation result	Recommendation succeeds	Generate recommendation; inspect history	Valid profile	Recommendation result/model version/timestamp is stored if history is enabled	Not executed — populate after manual test	TBD
IT_REC_06	IT	Handle ML service failure	ML service unavailable	Request recommendations	Valid profile	Backend returns controlled error/fallback response	Not executed — populate after manual test	TBD
ST_REC_07	ST	Show ranked career recommendations in UI	Valid recommendation response exists	Open recommendation screen	Sample result set	Career paths are displayed in descending relevance with scores/explanations	Not executed — populate after manual test	TBD
ST_REC_08	ST	Prevent recommendation without authentication	No active session	Open recommendation endpoint/page	Unauthenticated request	Request is rejected/redirected	Not executed — populate after manual test	TBD

GAP: Skill Gap Analysis — Owner: PES2UG24AM210 — Requirements: FR-19 to FR-24

ID	Type	Test Case Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_GAP_01	UT	Compute missing skills for selected career	Career has required skills; user has known skills	Run gap calculation	Target=Data Analyst; current=Python, SQL	Missing skills are identified correctly	Not executed — populate after manual test	TBD
UT_GAP_02	UT	Return empty gap when all required skills are present	User skills cover target requirements	Run gap calculation	All target skills present	No missing skills are returned	Not executed — populate after manual test	TBD
UT_GAP_03	UT	Normalize skill names before comparison	Equivalent names exist	Run comparison	Machine Learning vs machine-learning	Equivalent skill is matched	Not executed — populate after manual test	TBD
IT_GAP_04	IT	Retrieve target career requirements	Career exists in database	Open skill-gap page	Target career=Data Analyst	Required skill list is fetched	Not executed — populate after manual test	TBD

ID	Type	Test Case Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
IT_GAP_05	IT	Combine stored user skills with career requirements	User profile and career data exist	Run analysis	Valid profile + career	Gap result reflects both datasets	Not executed — populate after manual test	TBD
IT_GAP_06	IT	Handle unknown target career	Career ID is invalid	Run gap calculation	career_id=99999	Controlled not-found response is returned	Not executed — populate after manual test	TBD
ST_GAP_07	ST	Display required, possessed and missing skills	Gap calculation succeeds	Open gap results	Target career	UI clearly separates possessed and missing skills	Not executed — populate after manual test	TBD
ST_GAP_08	ST	Refresh gap after skill update	User adds a missing skill	Return to gap analysis	Added skill=Statistics	Previously missing skill is removed from gap	Not executed — populate after manual test	TBD

ROAD: Learning Roadmap — Owner: PES2UG24AM210 — Requirements: FR-25 to FR-28

ID	Type	Test Case Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_ROAD_01	UT	Generate roadmap for a valid skill gap	Skill gap exists	Generate roadmap	Missing: Statistics, Power BI	Ordered learning steps are returned	Not executed — populate after manual test	TBD
UT_ROAD_02	UT	Order prerequisites before dependent skills	Skills have prerequisite relation	Generate roadmap	SQL before advanced analytics	Prerequisite appears earlier	Not executed — populate after manual test	TBD
UT_ROAD_03	UT	Reject roadmap generation with no target career	No career selected	Click generate	No target career	System requests career selection	Not executed — populate after manual test	TBD
IT_ROAD_04	IT	Create roadmap from gap results	Gap service returns missing skills	Generate roadmap	Gap list	Roadmap contains missing skills as learning objectives	Not executed — populate after manual test	TBD
IT_ROAD_05	IT	Attach learning resource to roadmap step	Resource catalog contains item	Generate/open roadmap	Resource ID=R101	Resource title/link/type are available	Not executed — populate after manual test	TBD
IT_ROAD_06	IT	Handle unavailable resource	Resource is missing	Open roadmap step	Invalid resource ID	Step remains usable and missing resource is flagged	Not executed — populate after manual test	TBD
ST_ROAD_07	ST	Display roadmap in ordered stages	Roadmap exists	Open roadmap	Valid roadmap	Stages are shown in sequence with completion state	Not executed — populate after manual test	TBD
ST_ROAD_08	ST	Regenerate roadmap after profile/skill changes	User skill set changed	Regenerate roadmap	Updated skills	Roadmap reflects new gaps and does not retain obsolete required steps	Not executed — populate after manual test	TBD

PROG: Progress Tracking — Owner: PES2UG24AM214 — Requirements: FR-29 to FR-33

ID	Type	Test Case Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_PROG_01	UT	Mark roadmap item complete	Roadmap item exists	Click complete	Step=SQL Basics	Completion status becomes complete	Not executed — populate after manual test	TBD
UT_PROG_02	UT	Prevent invalid progress percentage	Roadmap item exists	Submit progress outside 0–100	progress=150	Validation rejects value	Not executed — populate after manual test	TBD
UT_PROG_03	UT	Update progress percentage	Roadmap item exists	Set progress to 50%	progress=50	Stored progress becomes 50%	Not executed — populate after manual test	TBD
IT_PROG_04	IT	Persist progress update	User and roadmap exist	Update progress; reload	Step=SQL, 50%	Progress remains 50%	Not executed — populate after manual test	TBD

ID	Type	Test Case Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
IT_PROG_05	IT	Recalculate overall roadmap completion	Multiple roadmap items exist	Complete/update one item	3/5 items complete	Overall completion reflects current item states	Not executed — populate after manual test	TBD
IT_PROG_06	IT	Use performance/progress data for recommendation refresh	Progress data exists	Trigger recommendation refresh	Low progress in selected area	Refresh service receives current progress data	Not executed — populate after manual test	TBD
ST_PROG_07	ST	Display progress dashboard accurately	Progress records exist	Open dashboard	Mixed completion states	Charts/percentages match stored records	Not executed — populate after manual test	TBD
ST_PROG_08	ST	Prevent another user from changing progress	Two accounts exist	Submit progress using another user's ID	User A token + User B item	Unauthorized update is rejected	Not executed — populate after manual test	TBD

11. Test Summary Template

Metric	Value
Total planned test cases	56
Unit tests	21
Integration tests	21
System tests	14
Passed	TBD
Failed	TBD
Blocked	TBD
Pass percentage	TBD
Open critical/high defects	TBD

12. Requirement Traceability Mapping

Functional Area	SRS Requirement Group	Primary Test IDs
Authentication & Account	FR-01 to FR-05	UT/IT/ST AUTH 01–08
Student Profile	FR-06 to FR-09	UT/IT/ST PROF 01–08
Skill Management	FR-10 to FR-13	UT/IT/ST SKILL 01–08
Career Recommendation	FR-14 to FR-18	UT/IT/ST REC 01–08
Skill Gap Analysis	FR-19 to FR-24	UT/IT/ST GAP 01–08
Learning Roadmap	FR-25 to FR-28	UT/IT/ST ROAD 01–08
Progress Tracking	FR-29 to FR-33	UT/IT/ST PROG 01–08

13. Team Ownership and Sign-off

Team Member / SRN	Primary Functionality	Signature / Date
PES2UG24AM184	Authentication + Student Profile	TBD
PES2UG24AM217	Skill Management + Career Recommendation	TBD
PES2UG24AM210	Skill Gap Analysis + Learning Roadmap	TBD
PES2UG24AM214	Progress Tracking + Platform Integration	TBD

Team Member / SRN	Primary Functionality	Signature / Date
Faculty	Dr. Siba Ram Baral / as applicable	TBD